

OVERLAP WITH AUTODESIGN

AUTODESIGN VS PROJET COMPLEXE (META-HARNESS OPTIMIZATION)

Date: 2026-08-24

Status: architecture note / design instrument (not a spec, not an implementation plan)

Reads: [Yaxin9Luo/AutoDesign](#), [arXiv:2608.13560](#), [autodesign.designanything.ai](#), ASC README, August 2026 ideas notes (Revival v2–v4, Reverse Prompting vs CLR, Meadows leverage, Four layers, Cognitive institutions), and the sibling note [Overlap with EnvHarness](#)

Paper dated 13 Aug 2026 (cs.CV / cs.AI / cs.CL). Code released 14 Aug 2026. Same week as EnvHarness (20 Aug 2026): two papers, two sides of the same “harness, not weights” claim. EnvHarness wraps a frozen *environment*. AutoDesign evolves the *agent* harness around a frozen *model*. Projet Complexe sits between them: named computational glue (ASC), packing and composition (Projet Complexe ASC), a living knowledge world (Projet Complexe).

VERDICT

Strong overlap on one axis, useful on a second, weak on the rest. AutoDesign is the closest published operationalization of Revival v4’s claim that **the product is the loop around a frozen model**, not the model. It shows that reusable skill can accumulate in software — prompts, tools, validators, orchestration — without touching weights. Projet Complexe wants that same accumulation for a local, multilingual second brain, with YAML + DSL as source of truth and a human as the only commit of Claims. Steal the two-loop picture, the five-component taxonomy, the one-component gated update, and the dual-critic split. Do not import their poster mill, their code-agent-rewrites-the-harness outer loop, or their VLM-as-judge as truth.

Dimension	Strength	Overlap	AutoDesign	Projet Complexe
Frozen model, evolve the surround	High	Strong overlap — one axis	DesignHarness around π_θ	<code>pre_llm</code> / <code>post_llm</code> , allowlisted entry points
Nested loops (artifact vs system)	High	Strong as a control picture	inner designer-critic; outer meta-harness	inner: pack / act / critique one Task; outer: evolve packs, skills, hooks — with HITL
Five-way harness cut	Med-high	Partial	Context, Tools, Runtime, Orchestration, Evaluation	v4 five layers + planes A–E; do not collapse
Dual critic (rules + perception)	Med	Partial	blocking checks + VLM critic	DSL <code>validate</code> : + HITL; not a vision judge as ontology
Product and ontology	Low	Weak on the rest	paper → poster / slides / web / video	ingest, Claims, Links, gaps, fr/en/pt, modest hardware
Self-rewriting production system	Conflict	Refuse if copied	coding agent patches harness source; train/dev gate	ASC core is not a Gödel machine; killswitch is System M, not an optimizer

WHAT AUTODESIGN ACTUALLY IS

Human communication often means turning long, heterogeneous sources into a short, structured artifact (poster, slides, page, video). That is a long-horizon agentic job: extract evidence, plan, render, take critique, repair. Existing design systems already run generate-critique-revise, but they treat each piece of feedback as **transient**. Humans accumulate priors. Static harnesses do not.

AutoDesign’s move: **optimize the harness H , not the artifact y , and not the weights θ .**

$$y \sim H(\pi_\theta, x, c)$$

π_θ
stays frozen. is the multimodal source (a paper). is the design context (medium, constraints). is everything around the model: prompts, skills, tools, workspace, loop control, validators. The objective is expected artifact quality under a human-aligned evaluator :

$$H^\star = \arg \max_H J(H), \quad J(H) = \mathbb{E}[R_{\text{meta}}(y, x, c)]$$

They instantiate this on **academic paper-to-poster**, then ship the learned system as **DesignHarness**. They also introduce **PosterBench** (100-paper Main Track, five disciplines; PosterBench-mini, 10 papers) with a seven-dimension rubric.

Reported numbers (treat as the paper’s claims; see caveats below):

Claim	Figure
PosterBench Main Track (Claude Code + Claude 4.8)	78.32, +7.45 vs Claude Design (70.87)
Seven model / coding-agent configs, harness on vs off	mean 54.99 → 67.39 (+ 12.40 points ; the abstract’s “+12.4%” is a unit slip)
Per-config gain range	+5.01 (Claude 4.8) to +19.56 (DeepSeek V4 Pro) — weakest models gain most
One autonomous poster run	253 tool calls, 11 editing turns, ~40 min, under \$3
Cheap configuration	LongCat 2.0 ≈ 55.13 at ~\$0.27 / poster
System-blind human study (11 reviewers, 933 judgments)	Bradley–Terry 64.0% (95% interval 55.2–77.8%)
Evolution trace	7 days, 224 subagents, ≥123 outer iterations, 54 accepted harness updates

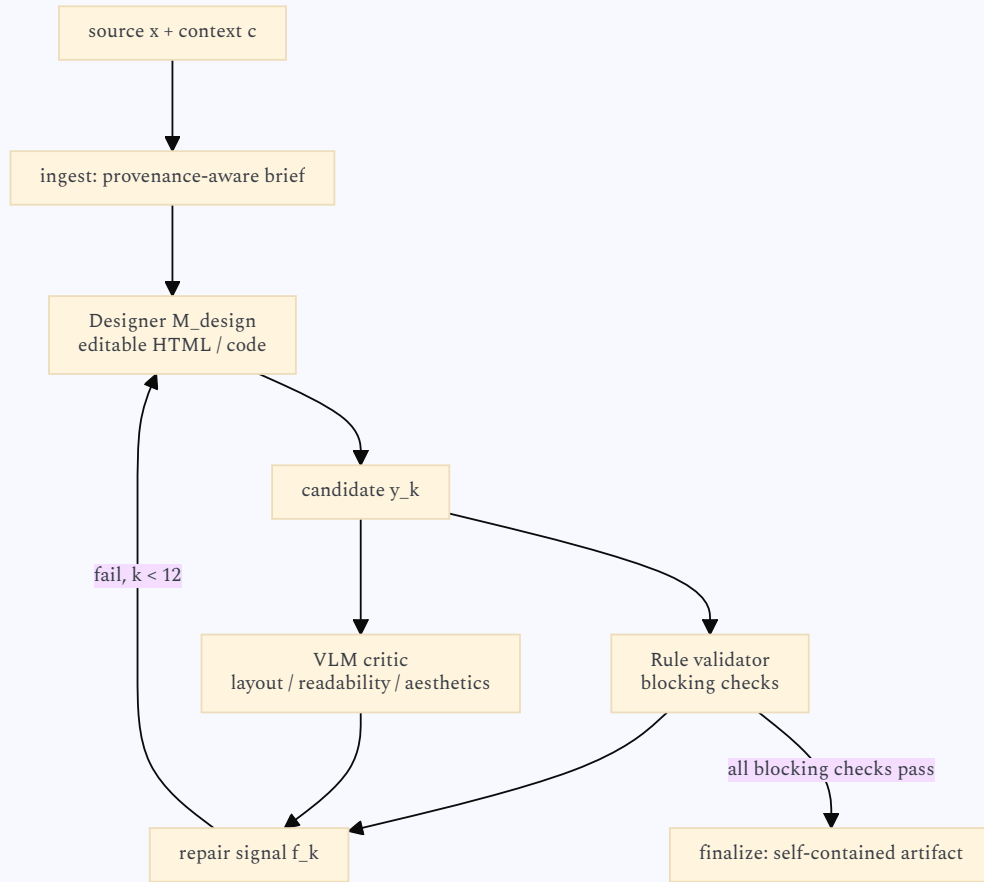
The repo ([Yaxin9Luo/AutoDesign](#)) ships DesignHarness, PosterBench records, a local workbench, and **standalone Agent Skills** (Poster, PPT, Webpage, Video) installable in Codex, Claude Code, DeepSeek Harness, Pi, OpenCode — without their application server.

TWO NESTED LOOPS

This is the diagram worth keeping. Tables cannot show the nesting.

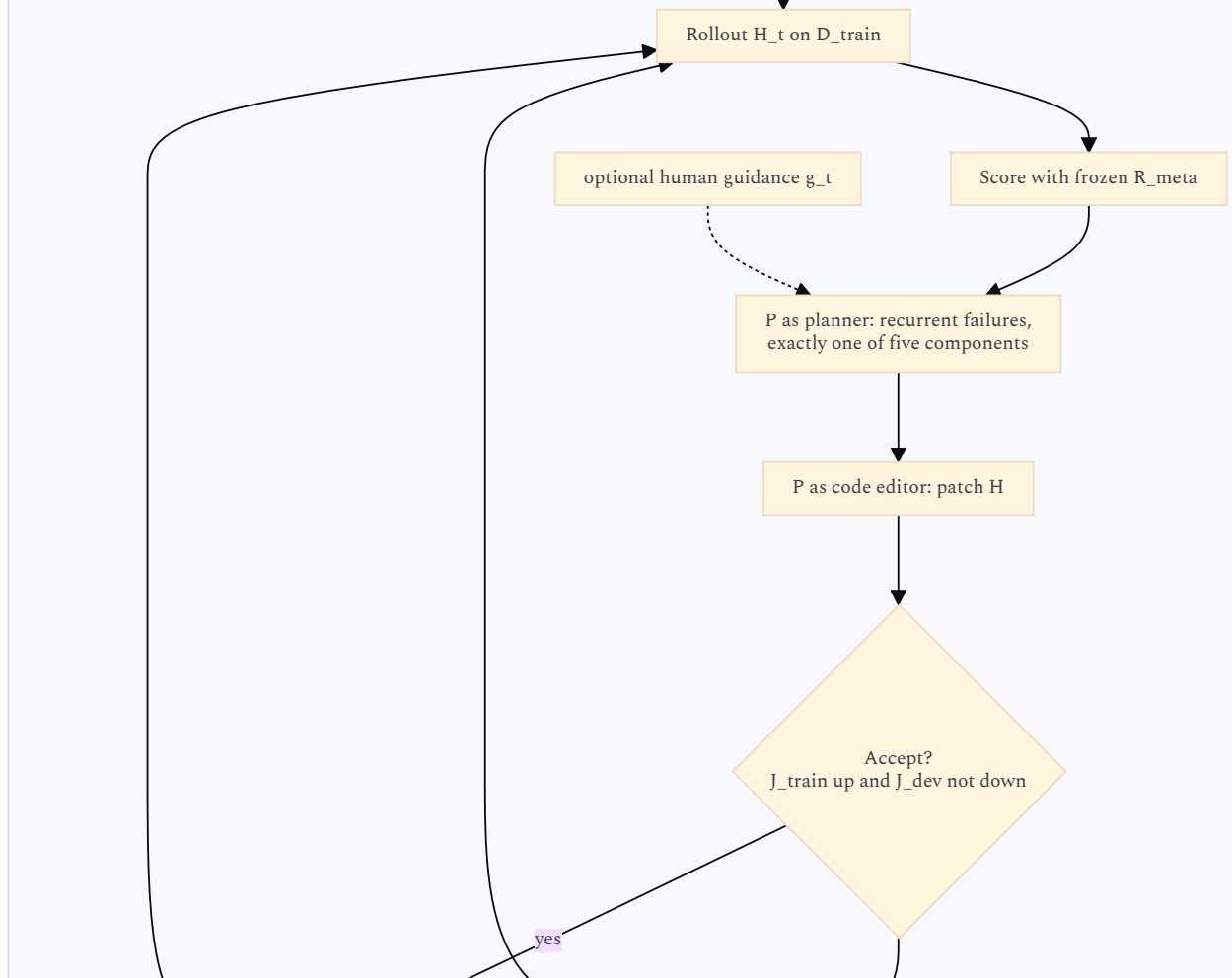
$xcHR_{\text{meta}}$

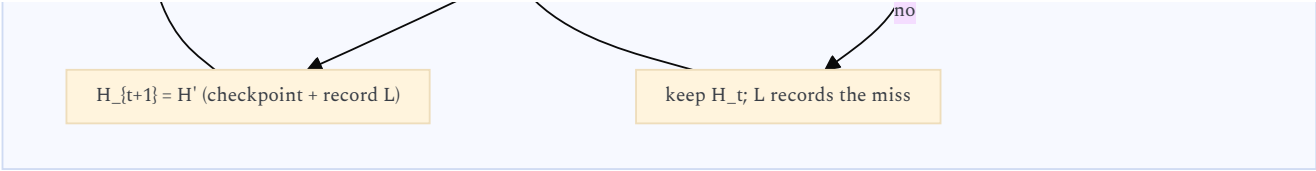
Inner loop — DesignHarness (fixed H, one artifact)



trajectories t , scores s

Outer loop — Meta-harness (updates H across tasks)





Inner loop improves *this* poster. H does not change. Designer and critic:

$$y_k = M_{\text{design}}(y_{k-1}, f_{k-1}; x, c), \quad f_k = M_{\text{critic}}(y_k; x, c)$$

Artifact stays **editable HTML** (localized patches, not full regeneration). Blocking checks can stop the loop early. Budget $K = 12$. Fallback if nothing passes.

Outer loop improves H across tasks. Four stages: rollout, evaluation, update proposal, acceptance. Optimizer P is a coding agent that first plans, then edits. Each iteration may touch **exactly one** of the five components (several files inside that component are allowed). Development scores are used **only** by the gate and are **hidden from P** , so $\mathcal{D}_{\text{dev}}$ is an overfitting guard, not extra training signal. Record $\mathcal{L}$ stores harness checkpoint, traces, chosen component, plan, diff, accept/reject. No tree search: one active harness at a time.

Optional HITL: natural-language g_t to the planner when the outer loop plateaus; or human correction of R_{meta} if a systematic bias shows up. Humans do not edit harness code directly. With no guidance, the outer loop runs autonomously.

That split — **repair the artifact vs repair the system that will produce the next artifact** — is the idea worth stealing as vocabulary. It is not a reason to let a coding agent rewrite ASC core.

FIVE COMPONENTS OF H

AutoDesign component	What it holds	Projet Complexe / ASC analogue	Steal?
Context and Memory	source management, prompts, skills, reusable assets, persistent state	packing (pre_llm), locale-aware retrieve, procedure files / SKILL.md, notes — not “memory = vector DB”	Map. Keep v3: index is a projection.
Tools and Specifications	tools + editable artifact specs (layout, typography, provenance)	allowlisted ASC entry points; YAML able → JSON Schema projection; DSL addressing	Map. YAML remains SoR.
Execution Runtime	workspace for authoring, rendering, validating, exporting	\$PROJECT_DOCROOT, Compose stack, Pi / Ollama as Technology — not a second coding TUI	Delegate runtime. Host Pi; do not become Pi.
Orchestration	routing, attempt budgets, loop control, candidate selection, fallback, finalization	PCA cascade / router (paper 2603.04445), killswitch, HITL, attempt limits	Map as policy , not a new protocol.
Evaluation and Feedback	rule-based validation, model-based critique, localized repair	DSL validate; Winteringham-style eval around traces and publish; HITL on Claims	Split: rules yes; VLM-as-truth no.

Revival v4 already has a **different** five-layer cut (Model / Prompt assembly / Tool surface / Transport / Procedure pack) plus planes A-E (routing, index, language, knowledge/fusion, orientation). AutoDesign’s five are a *functional* cut of the production system. The v4 five are a *interface* cut of how a model gets tools. Complementary. Do not merge them into one table and call it ontology.

v4 layer (how agents get tools)	AutoDesign component (what H is made of)	Relation
1. Model	frozen π_θ (outside H)	Same freeze
2. Prompt assembly	Context and Memory	Overlap
3. Tool surface	Tools and Specifications	Overlap
4. Transport (MCP, ...)	mostly absent; Skills as files	v4 already: MCP is a plug
5. Procedure pack	skills inside Context and Memory	They install SKILL.md in <i>other</i> harnesses — that is the v4 “emit, don’t become” policy in the wild
—	Execution Runtime	v4 delegates this
—	Orchestration	v4 plane A + E (router + System M)
—	Evaluation and Feedback	v4 post_llm + Winteringham

DESIGNHARNESS AS A PRODUCTION PIPELINE

After the outer loop, the learned system is a four-stage pipeline. This is closer to Projet Complexe’s *ingest* → *pack* → *act* → *export* than to EnvHarness’s *reset* / *step* wrap.

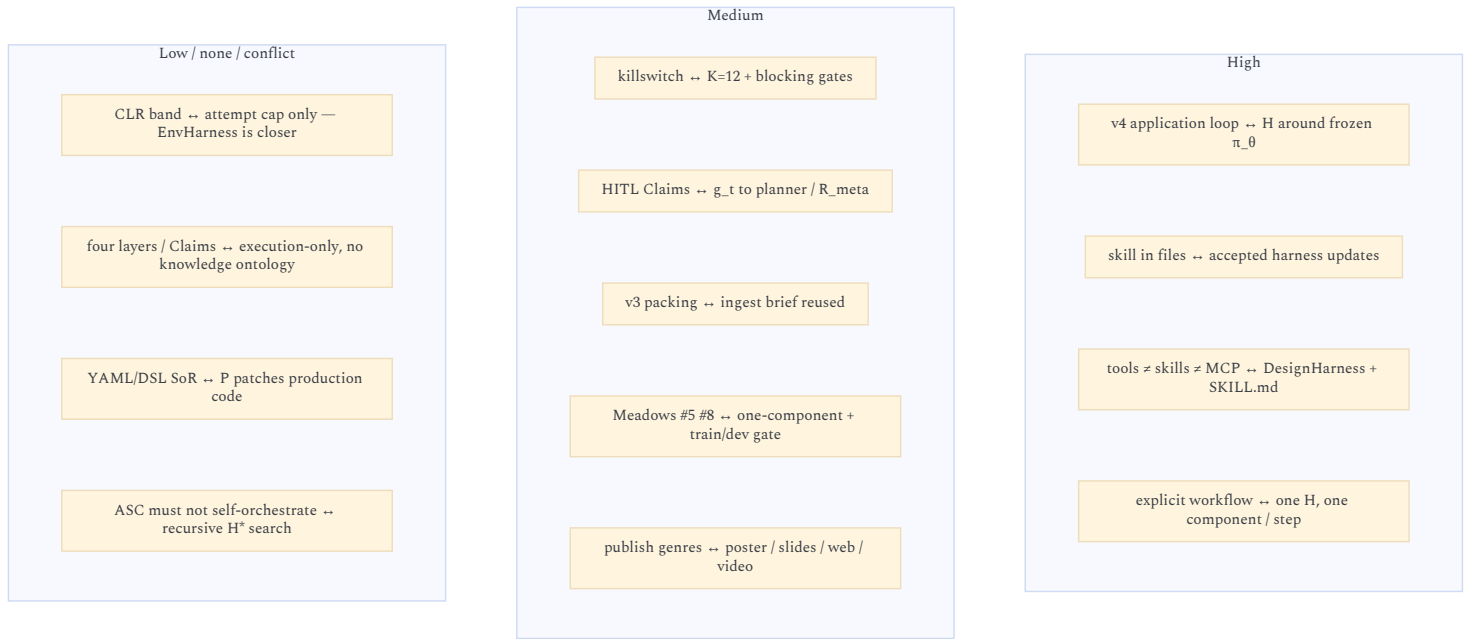
Stage	What it does	PC analogue	Risk if copied as identity
Paper ingestion	metadata, outline, claim-supporting passages, figures/tables with source locations ; content brief + medium plan	extract as capability; provenance on Claims; locale metadata	Becoming a CAT/poster product; GROBID/Docling as identity (v3 already refused that)
Generation and revision	coding-agent Designer; HTML stays editable; render PNG/PPTX/MP4 for critique	run-agent hosted in Pi; publish drafts as files	Coding agent as second brain
Validation	deterministic blocking checks (unsafe assets, broken provenance, overflow, typography) + VLM on render	DSL tests; killswitch; HITL	VLM critic as knowledge commit
Finalization	inline assets, typesetting, self-contained deliverable; fallback if budget exhausted	publish with a contract (post, chapter, note)	Prompt pack as publish (v4 refuse)

The blocking-check idea is the clean steal: **some failures must stop the loop**; others are non-blocking diagnostics (coverage, density, numeric consistency). That is already how ASC DSL **test-*** is meant to work: fail closed on contract, comment on taste.

CONCEPT MAP (NOTES → AUTODESIGN)

Same control problem as Revival v4 and the Flow note, different object being optimized, different ethics of “the system rewrites itself.”

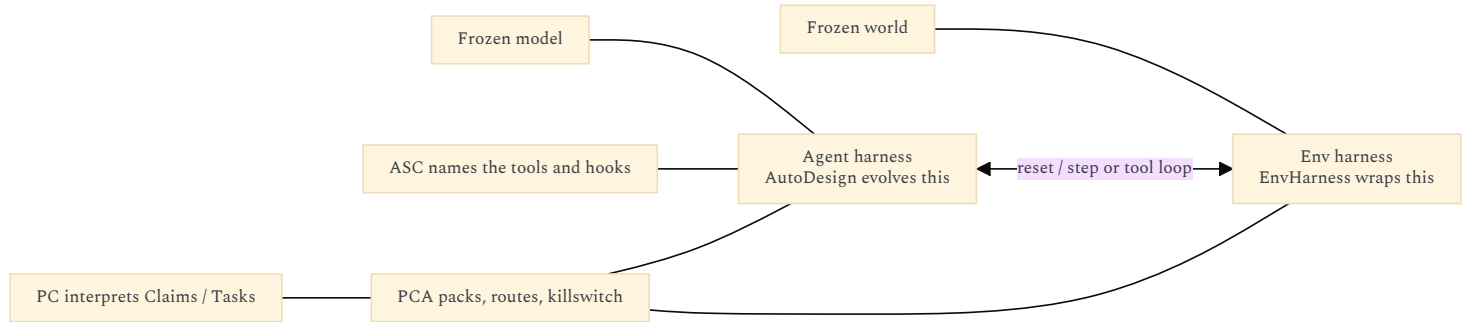
Idea in your notes	AutoDesign counterpart	Overlap
v4: the product is the application loop around a frozen model (pre_llm / post_llm , allowlisted tools)	H around frozen π_θ ; inner designer–critic is that loop made explicit	Highest. Cite this paper as the <i>agent-harness optimization</i> sibling of EnvHarness.
“Durable skill lives in files, not in chat logs” (v4 plane E; Dupoux/LeCun/Malik: do not fake System A with transcripts)	54 accepted updates accumulate in the harness repo; $\mathcal{L}$ is the optimization memory	High as a claim. Conflict on <i>who</i> writes the files: they let P patch production code; you want YAML/DSL authored or HITL-accepted.
Tools $\neq$ skills $\neq$ MCP	DesignHarness tools + standalone Agent Skills + no MCP-as-brain	High. They even ship Skills for Pi / Claude Code / Codex. That is the v4 “export SKILL.md , do not make ASC a skill runner.”
Explicit workflow default (Dibia); refuse swarms	named stages; one active H ; one component per outer step	High as discipline. Credit assignment by construction.
Killswitch / System M (Dupoux et al. 2603.15381 ; v2 mutual killswitch)	attempt budget $K = 12$; blocking gates; optional g_i	Medium. Their M is a <i>budget and a gate</i> . Yours is <i>task vs research vs stop</i> , and it can kill research when the Task says so.
HITL commit of Claims; never multi-agent consensus as truth (2603.24402)	human guidance to planner; human may revise R_{meta} ; R_{meta} otherwise frozen	Medium. Directional HITL on the <i>optimizer</i> is close. HITL on <i>knowledge</i> is not what they do.
v3 RAG as context regulation so a 7B stays in the Flow band	ingestion brief constructed once, reused across inner steps; packing is Context and Memory	Medium. Same intent (fit the window with a structured substrate). They pack a paper for a poster; you pack a corpus for a Claim.
Flow / CLR: keep complexity in a band	$K = 12$, blocking vs non-blocking, fallback	Medium–low. They cap <i>horizon</i> . They do not measure a CLR band or scaffold vs harden the <i>task</i> . EnvHarness is closer on that axis.
Meadows #6 information flows; #5 rules; #8 balancing feedback	who sees traces (P sees train, not dev); which edits are legal (one component); write-and-validate gate	Medium–high on #5 and #8 for the <i>harness</i> . Not for the knowledge graph.
Four layers (ontology / semantics / dynamics / execution)	almost entirely execution , with a thin dynamics loop (outer H updates)	Low as ontology. No account of what a Claim is.
Three-project cut; genericity scale; killswitch as identity	one Python/TS workbench + Skills	Low. Different product.
Knowledge plane: Claims, Links, gaps, provenance, fr/en/pt	provenance <i>links back to the PDF</i> for poster fidelity — not a curated graph	None as ontology. Do not collapse “source location on a figure” with a Claim.
YAML entity/able + DSL as SoR; JSON Schema / MCP as projections	P emits real patches to harness source; Skills are markdown+scripts	Conflict if copied. Same fight as EnvRigger’s Python subclasses.
ASC non-goal: “self-organizing all-orchestrating platform”	recursive meta-harness improvement is the headline	Conflict. Steal the <i>gated, one-component</i> recipe for PCA packs. Refuse it as ASC core behavior.
publish as versioned export (notes, posts, book chapters) with quality contracts (Yu & Yao)	paper → poster / slides / web / video as media	Medium as an export family. One publish Implementation among others — not the product.



AUTODESIGN, ENVHARNESS, PROJET COMPLEXE (THREE-WAY)

Do not treat the two August papers as competitors. They freeze different things.

	EnvHarness (2608.19880)	AutoDesign (2608.13560)	Projet Complexe
Frozen core	environment + human verifier	model weights θ	CLIs / OS / named pivots (ASC); human goal / Claim commit (PC)
What gets rewritten	Stage / Contract / Chain wrappers	five components of H	packing, allowlists, compositions, procedure files — not core names, not the user's objective
Inner loop	policy rollout in wrapped env	designer-critic on one artifact	one Task: retrieve, act, critique, HITL
Outer loop	EnvRigger writes env wrappers from diagnosed policy flaws	P writes harness patches from recurrent failures	<i>optional later:</i> evolve PCA packs/skills under a gate — never ASC core
Success metric	held-out benchmark SR / steps	PosterBench + blind preference	ingest, curate, export, research; same named actions for human and agent; fr/en/pt
SoR for the wrap	Python subclasses	harness source (Python/TS) + Skills	YAML + DSL; JSON Schema / MCP / SKILL.md as projections
Ethics of “make it fit”	hide the mug (legitimate on a benchmark)	change the <i>procedure</i> , keep the paper	never silently rewrite the user's task or mint a Claim without HITL



EnvHarness answers: *reshape what the agent sees and may do*. AutoDesign answers: *reshape the software that will produce the next artifact*. Projet Complexe needs both answers as **policy**, and neither as **product identity**.

STEAL

Cite AutoDesign as the agent-side of the harness idea Revival v4 already split from MCP / skills / tools, and as the counterpart to EnvHarness's env-side. Name **two loops** in PCA policy: (1) inner — pack, act, dual-check, stop; (2) outer — only if you later evolve packs/skills, and only behind HITL + a frozen eval. Name **five H parts** when talking about **run-agent**, without renaming ASC subjects after them. Map: packing $\rightarrow$ Context; allowlisted pivots $\rightarrow$ Tools; Pi/Ollama/Compose $\rightarrow$ Runtime (delegated); router + killswitch $\rightarrow$ Orchestration; DSL tests + HITL $\rightarrow$ Evaluation. Steal **one-component, train-up / dev-not-down**. If PCA ever auto-edits a pack, that is the whole safety recipe. Credit assignment stays readable. $\mathcal{D}_{dev}$ hidden from the proposer is the important detail.

Steal **blocking vs non-blocking** checks. Contracts fail closed. Taste does not.

Steal **Skills as interchange**. Emit `SKILL.md` for Pi/Claude; keep executables as ASC entry points. They already did the “works in the host you already have” version of v4 §3.3.

Steal **editable artifact + localized repair**. `publish` drafts should be files you patch, not a regenerated blob.

Steal **keep the final evaluator frozen and separate** from the optimizer’s R_{meta} . That is the same discipline as EnvHarness keeping the human verifier frozen. Apply it to Claim quality vs packing heuristics.

Steal the **empirical hint** that harness gains are largest on weaker models. That supports modest-hardware routing (plane A): a local 7B plus a good pack can move more than a bigger model plus a stuffed window — which is already the README Flow tease.

REFUSE

Do not put a MetaHarnessOptimizer, EnvRigger-class designer, or GRPO in ASC core. Complex agent/NL work stays in dedicated instances (ASC README non-goals).

Do not let a coding agent patch `asc/` or rewrite pivots because “the outer loop accepted it.” Genericity scale and HITL exist so names do not drift under automated taste.

Do not make PosterBench, Claude Design, or “conference-poster quality” Projet Complexe’s success metric. Yours is ingest, curate, export, research, three languages, modest hardware, same named actions for human and agent.

Do not treat DesignHarness as the second brain, or the workbench UI as Projet Complexe. Wrap a harness; do not become a design studio.

Do not use a VLM critic as the commit of a Claim. Dual critic is fine when the blocking side is deterministic and the perceptual side is advisory. Knowledge fusion stays Koch-style uncertainty + HITL, not “the poster looked good.”

Do not start with slides/web/video “same harness, more media.” Even they call those pilots. `publish` genres are contracts (Yu & Yao), each with its own evaluator — which is also their own §6 warning.

Do not copy R_{meta} and PosterBench sharing the same seven-dimension vocabulary as a pattern for “research quality.” The paper asserts the functions are distinct; a referee would still demand a correlation check. For Claims, keep the human as the only SoR.

CAVEATS ON THEIR NUMBERS (SO YOU DO NOT OVER-CITE)

These do not kill the *idea*. They bound how hard you lean on the *score*.

Issue	Why it matters here
The paper never states that $\mathcal{D}_{\text{train}} / \mathcal{D}_{\text{dev}}$ are disjoint from PosterBench Main Track / mini	78.32 and +12.4 points are unbiased only under an unstated assumption. Cite the <i>recipe</i> , not the headline, until that split is public.
R_{meta} and PosterBench share seven dimension names and the same authors	Optimizing H toward R_{meta} may secretly optimize toward PosterBench. Same trap as using RAGAS as both trainer and judge.
Abstract “+12.4%” vs body “+12.40 points”	Unit error; use points.
Human “conference-poster quality”	The study is pairwise preference among systems, not rating against real conference posters. Krippendorff α (nominal) is 0.101. BT interval (55.2–77.8%) overlaps Claude Code’s point estimate.
Main Track 78.32 has no uncertainty interval	N=100; still want a bootstrap before treating +7.45 as a wall.

WHERE THE ANALOGY BREAKS

AutoDesign is a **training-and-production curriculum for a design procedure**. It mutates the *factory* until posters score well, then runs that factory on a new paper. Projet Complexe is a **runtime cognitive institution**: a human-judged knowledge world plus a named computational vocabulary.

That changes the ethics of “the system improves itself.” In their setting, patching the poster validator after recurrent overflow is the point. In a second brain, an outer loop that rewrites `relate` or the Claim schema because a VLM preferred denser cards is a silent ontology change. Legal outer-loop objects on your side are **packs, skill files, packing heuristics, allowlists** — Meadows information flows and rules — not the primordial YAML, not the user’s Task, not a published Claim.

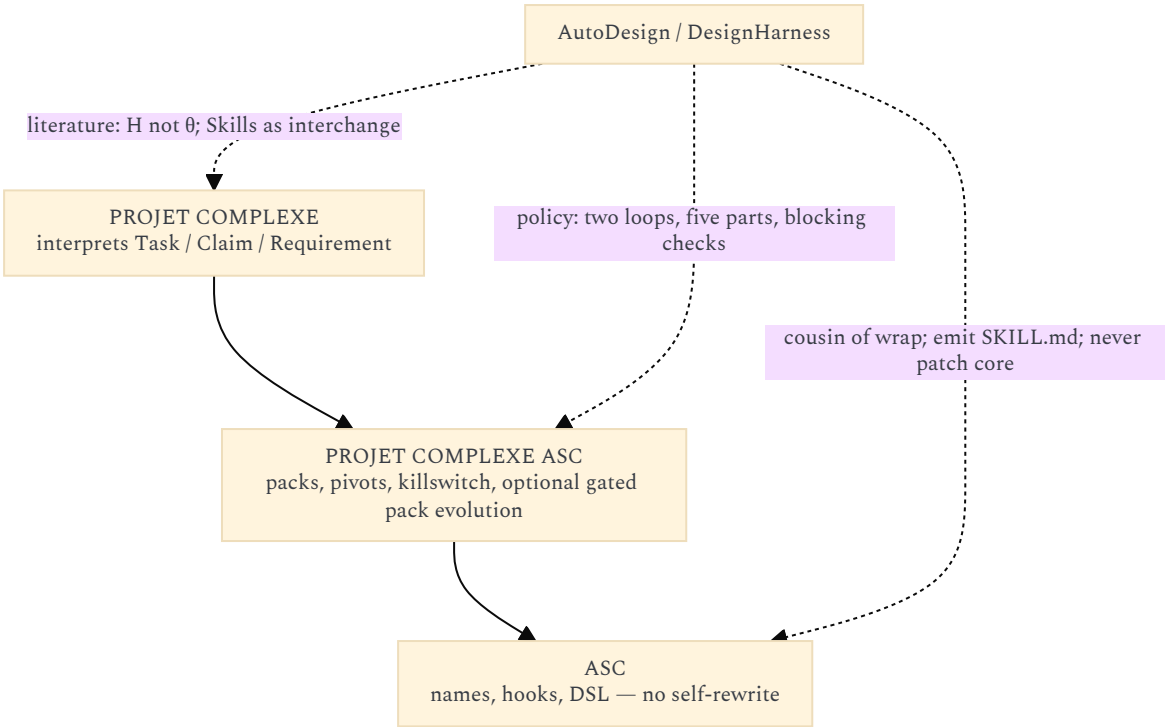
Four-layer note: AutoDesign lives in **execution**, with a thin **dynamics** loop (54 updates to H). Ingestion “claims” are passages extracted for a poster, not Claims in the knowledge plane. Cognitive institutions note: they improve the *procedure that produces artifacts*; Superpowers-like skills improve the *agent’s process pack*; Guardrails constrain I/O. AutoDesign is closest to the first, ships the second as `SKILL.md`, and uses blocking checks as a thin third. It is none of Projet Complexe’s knowledge institution, and complementary if kept in the export/worker slot.

Flow / CLR: they regulate **attempt count and blocking difficulty**, not the agent’s operating band. If a 7B is under-challenged on “rename this file,” AutoDesign will still run a twelve-step designer-critic unless Orchestration says otherwise. EnvHarness’s [0.4, 0.6] SR band is the closer measurement cousin of CLR. AutoDesign’s cousin is **horizon and contract**, not challenge calibration.

PLACEMENT ON THE THREE-PROJECT CUT

Project	Relation to AutoDesign
ASC	Already the more general wrap-don’t-rebuild layer (hooks, pivots, specificity). Document “harness = system around a frozen model” as a cousin of “thin wrapper over a frozen implementation.” Do not add a self-rewriting outer loop or a gym/design ABC. Optional: generate <code>SKILL.md</code> / JSON Schema from YAML <code>able</code> so other hosts can load the same procedures.

Project	Relation to AutoDesign
Projet Complexe ASC	If anything is stolen, it is policy : <code>pre_llm</code> packing as Context; allowlisted pivots as Tools; Pi/Ollama as Runtime; cascade + killswitch + <i>K</i> as Orchestration; DSL blocking checks + HITL as Evaluation. An optional later outer loop may evolve packs and skill files with one-component diffs and a train/dev-style gate — Implementation, not protocol.
Projet Complexe	The Flow UI / second brain remains the interpreter. AutoDesign does not help Claims, Links, gaps, or fr/en/pt as first-class. It is a literature ally for the README harness tease, and a possible worker behind <code>publish</code> (poster/slides as genres), not a UI or store.



PRACTICAL TAKEAWAY

Read AutoDesign next to EnvHarness, not instead of it.

1. **EnvHarness** — freeze the world, wrap `reset/step`, keep the verifier, regulate difficulty.
2. **AutoDesign** — freeze the model, evolve *H*, keep a frozen *eval* protocol, accumulate procedure in files.
3. **Projet Complexe** — freeze the user’s objective and the ASC names; regulate packing and tool surface; accumulate skills as files; only a human commits Claims.

The overlapping sentence all three can share:

Capability that survives a model swap lives in the named surround — not in weights, not in a chat log.

Your surround is ASC + PCA packs. Theirs is DesignHarness. EnvHarness’s surround is the env wrap. Copy the sentence. Copy the gates. Do not copy the factory.

Sources: AutoDesign paper abstract + §§1–8, Algorithm 1, Tables 1–4 (Stage/Contract analogue: five components, inner/outer loops, DesignHarness, PosterBench, ablations, human study, related work on Meta-Harness / STOP / ADAS / GEPA / Robeyns self-improving coding agent); repo README (Skills, hosts, paper→four artifacts); ASC README (wrap, Flow tease, agent extension, non-goals); Revival v2–v4; Reverse Prompting vs CLR; Meadows leverage; Four layers; Cognitive institutions; Overlap with EnvHarness.

- [arXiv:2608.13560](#)
- [github.com/Yaxin9Luo/AutoDesign](#)
- [autodesign.designanything.ai](#)
- [designanything.ai](#) (demo)
- Sibling: [Overlap with EnvHarness](#)
- [projet-complexe](#)
- [projet-complexe-asc](#)